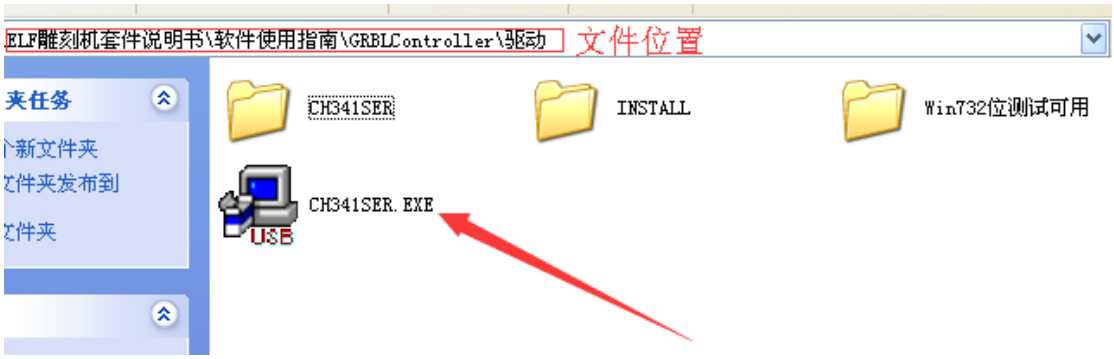
AX4 Pen Plotter Software Guide

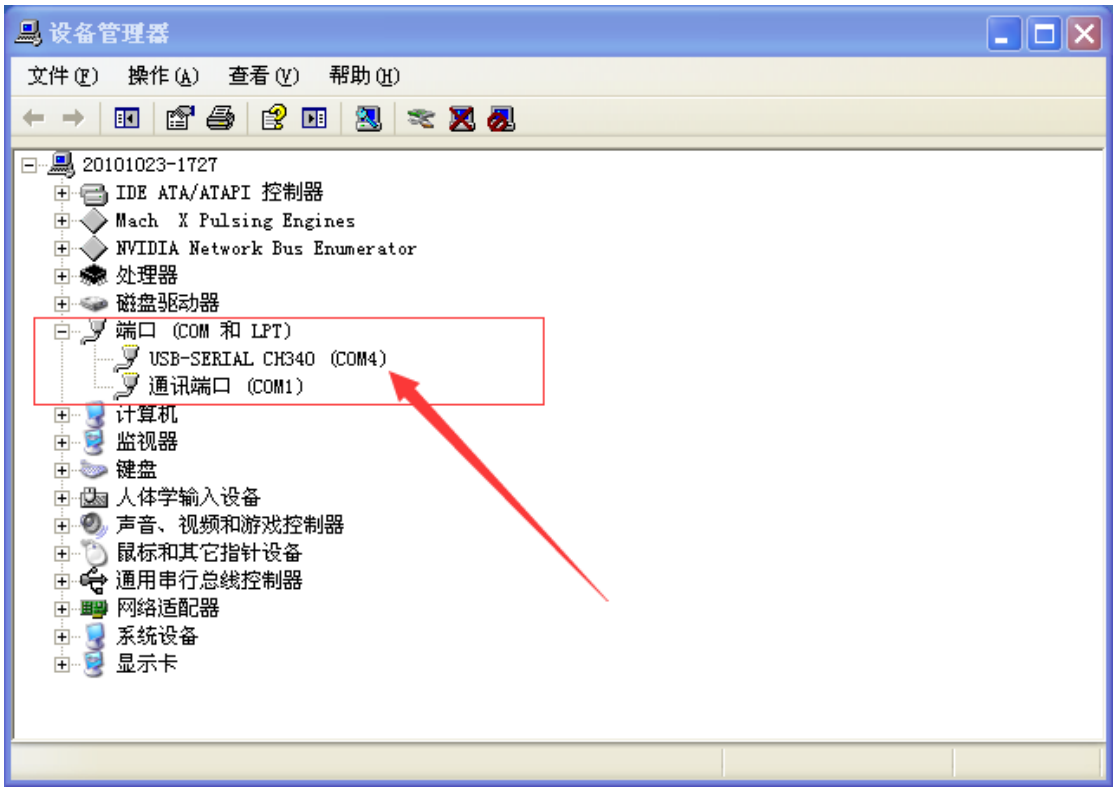
1、Install the driver

Connect the motherboard to the computer using a USB cable
Install driver software: Follow the path shown in the figure below to select the one that is suitable for your computer and install the driver



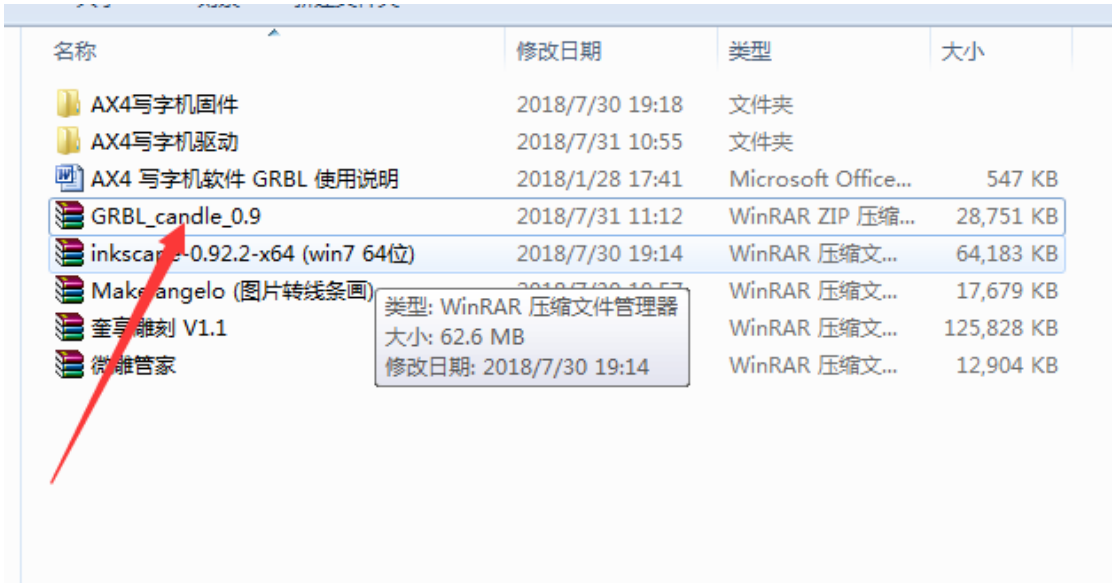


After installation, right-click My Computer>Manage>Device Manager>Port (COM and LTP)>View the port number



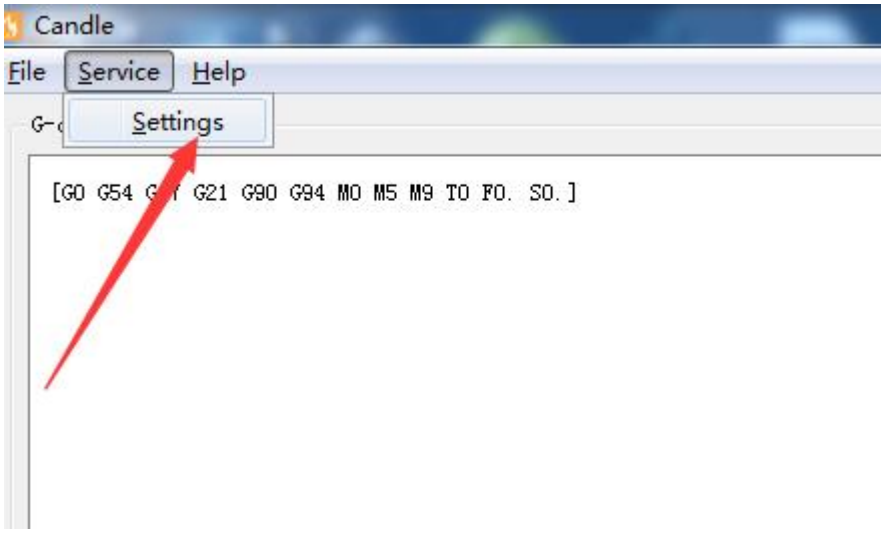
2、Install GRBL control software

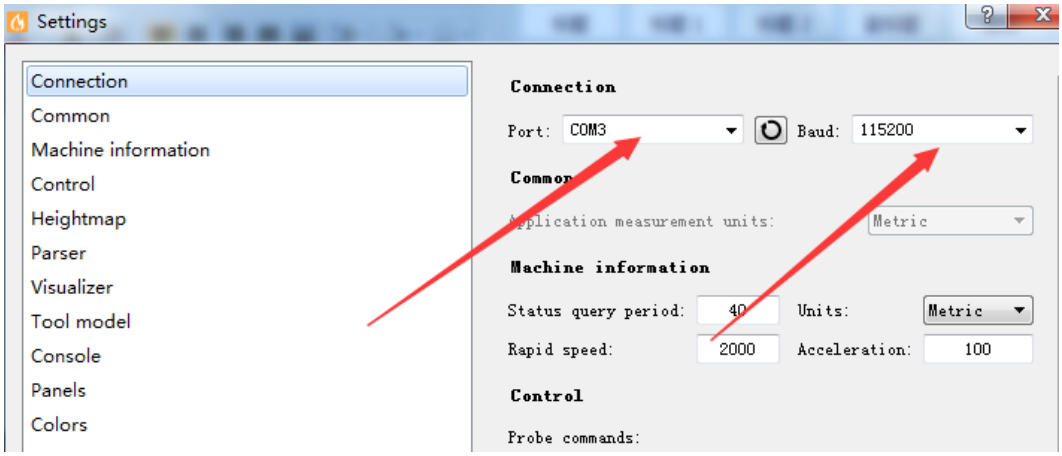
Select the appropriate version as needed, and we recommend GRBL_ candle_ zero point nine



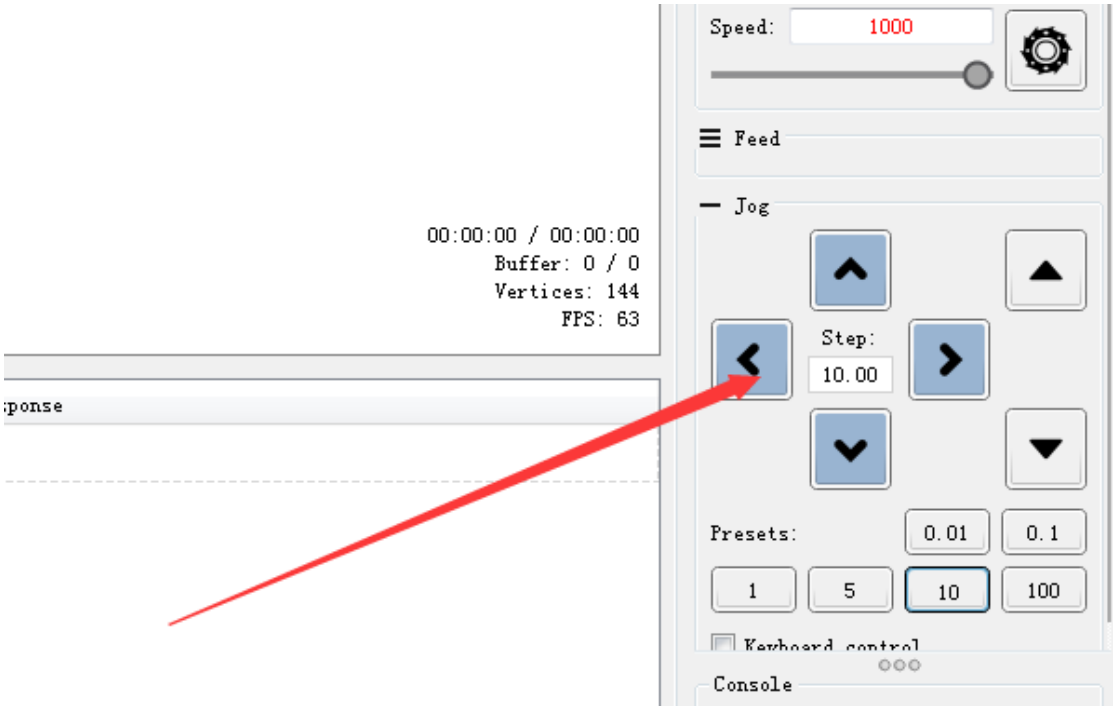
After installation, open the GRBL upper computer software, select Settings>Port>Baud rate 115200>, and then click Finish. After successful connection,

the direction button will turn dark.



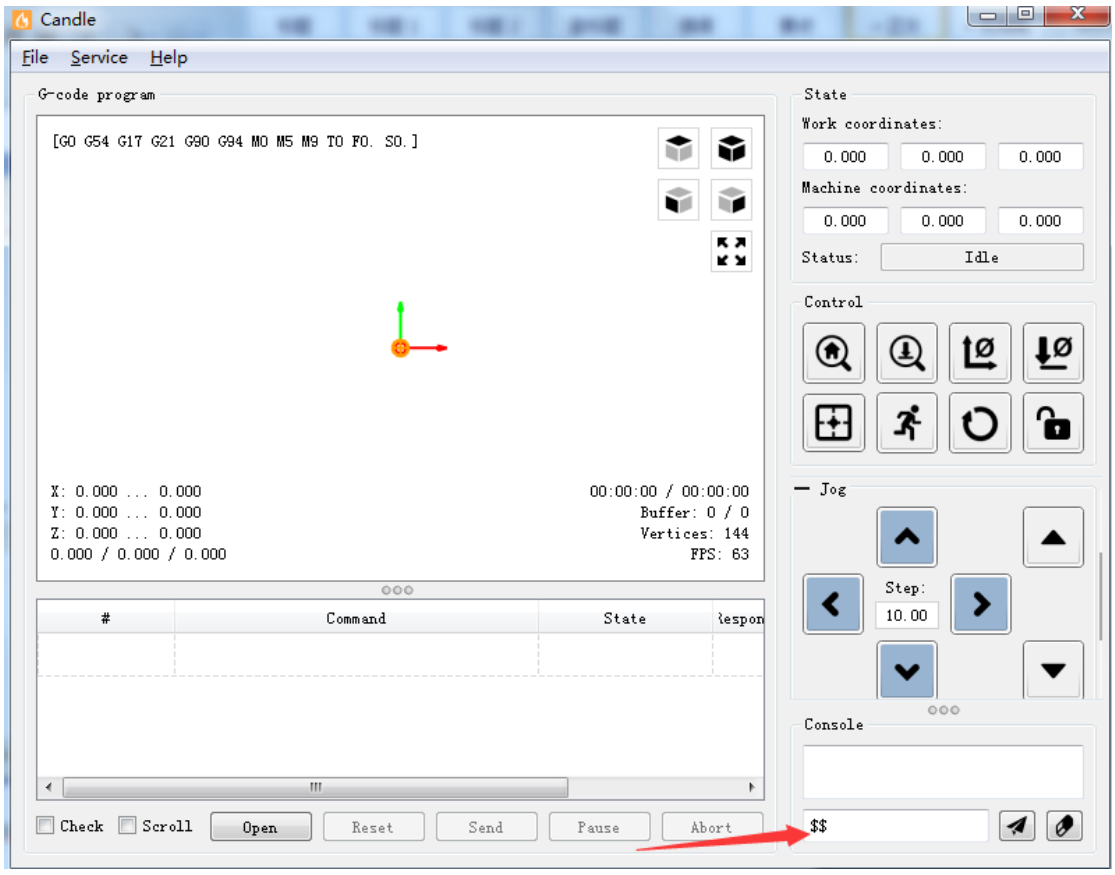


After connecting Grbl, the color inside the small box of the direction keys in the following image changes from gray to light
blue



3、GRBL control software settings

By default, we have burned the AX4 writing and drawing mode firmware for the customer, and various parameters have been set. The customer can send $$to the console to confirm the parameters。



```
[CTRL+X] < Grbl 0.9i ['$' for help]
$$ < $0=10 (step pulse, usec)
$1=25 (step idle delay, msec)
$2=0 (step port invert mask:00000000)
$3=0 (dir port invert mask:00000000)
$4=0 (step enable invert, bool)
$5=0 (limit pins invert, bool)
$6=0 (probe pin invert, bool)
$10=3 (status report mask:00000011)
$11=0.010 (junction deviation, mm)
$12=0.002 (arc tolerance, mm)
$13=0 (report inches, bool)
$20=0 (soft limits, bool)
$21=0 (hard limits, bool)
$22=0 (homing cycle, bool)
$23=0 (homing dir invert mask:00000000)
$24=25.000 (homing feed, mm/min)
$25=500.000 (homing seek, mm/min)
$26=250 (homing debounce, msec)
$27=1.000 (homing pull-off, mm)
$100=80.000 (x, step/mm)
$101=80.000 (y, step/mm)
$102=80.000 (z, step/mm)
$110=6000.000 (x max rate, mm/min)
$111=6000.000 (y max rate, mm/min)
$112=6000.000 (z max rate, mm/min)
$120=600.000 (x accel, mm/sec^2)
$121=600.000 (y accel, mm/sec^2)
$122=600.000 (z accel, mm/sec^2)
$130=300.000 (x max travel, mm)
$131=210.000 (y max travel, mm)
$132=200.000 (z max travel, mm)
Ok
```

If you need to make changes, you can send the corresponding code to the console. For example, if you want to modify the maximum speed of the X-axis to 1800, you can send $110=1800.000 to the console.

For more information, please refer to the official website:
Grbl v0.9 version notes
https://github.com/papabear113/grbl/wiki/Configuring-Grbl-v0.9

Common commands for GRBL:

View All Instructions

$X - Kill alarm lock unlock

$H - Run homing cycle zero return cycle, automatic zero return, provided the limit is opened

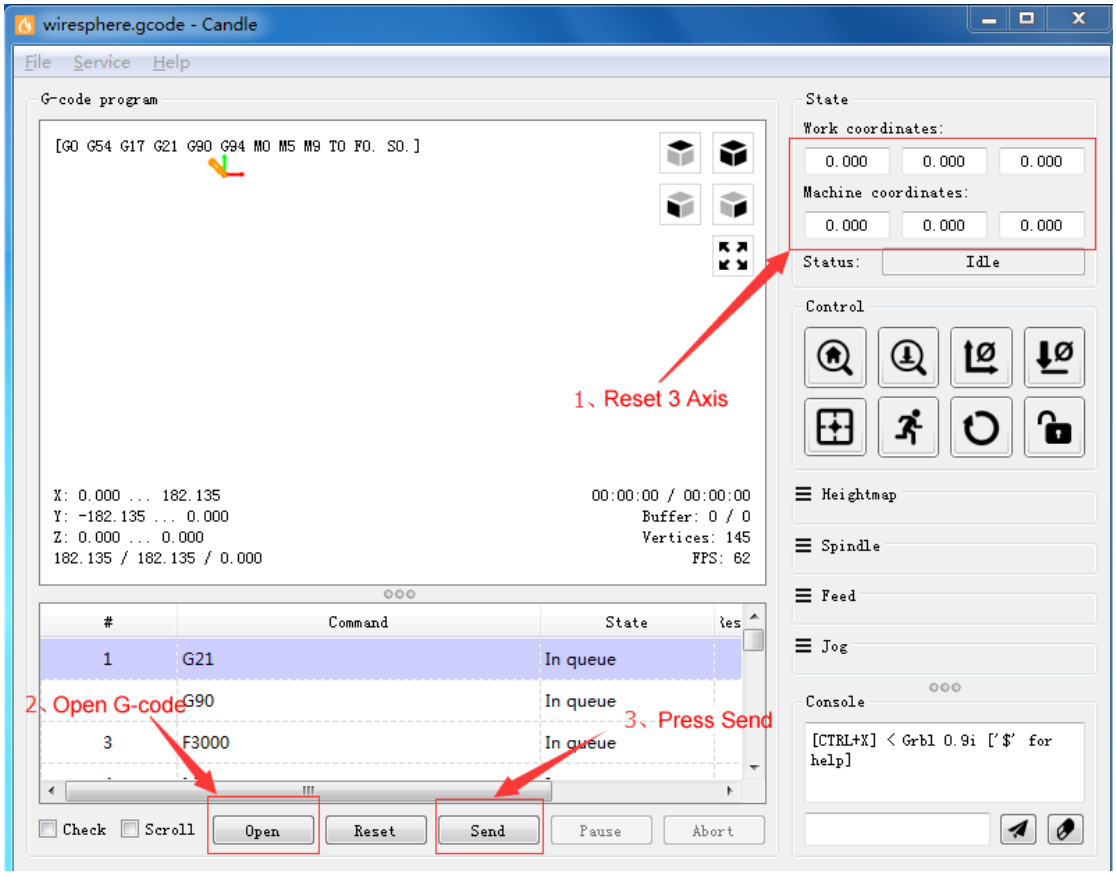
G92 X0 Y0 Z0- Set the current position to<x, y, z>=<0,0,0>

M03- Spindle forward rotation

M05- Spindle stop

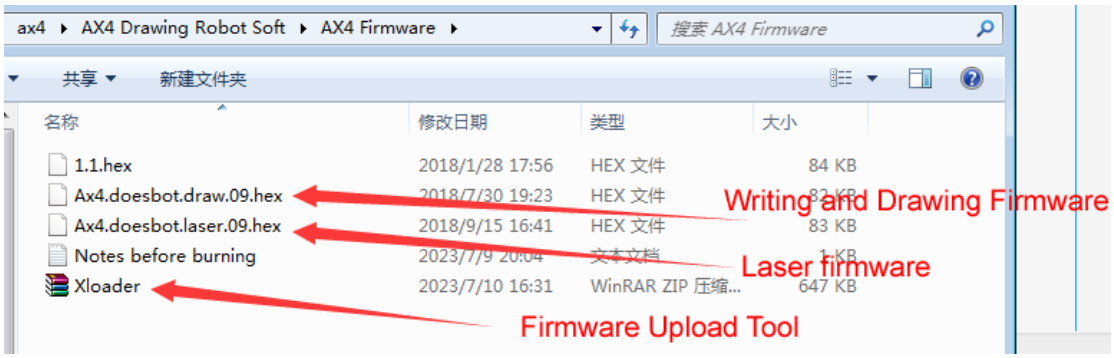
4、file run

Press the controller reset button first, reset the three axes, select Open File, select Send, and you can start carving.

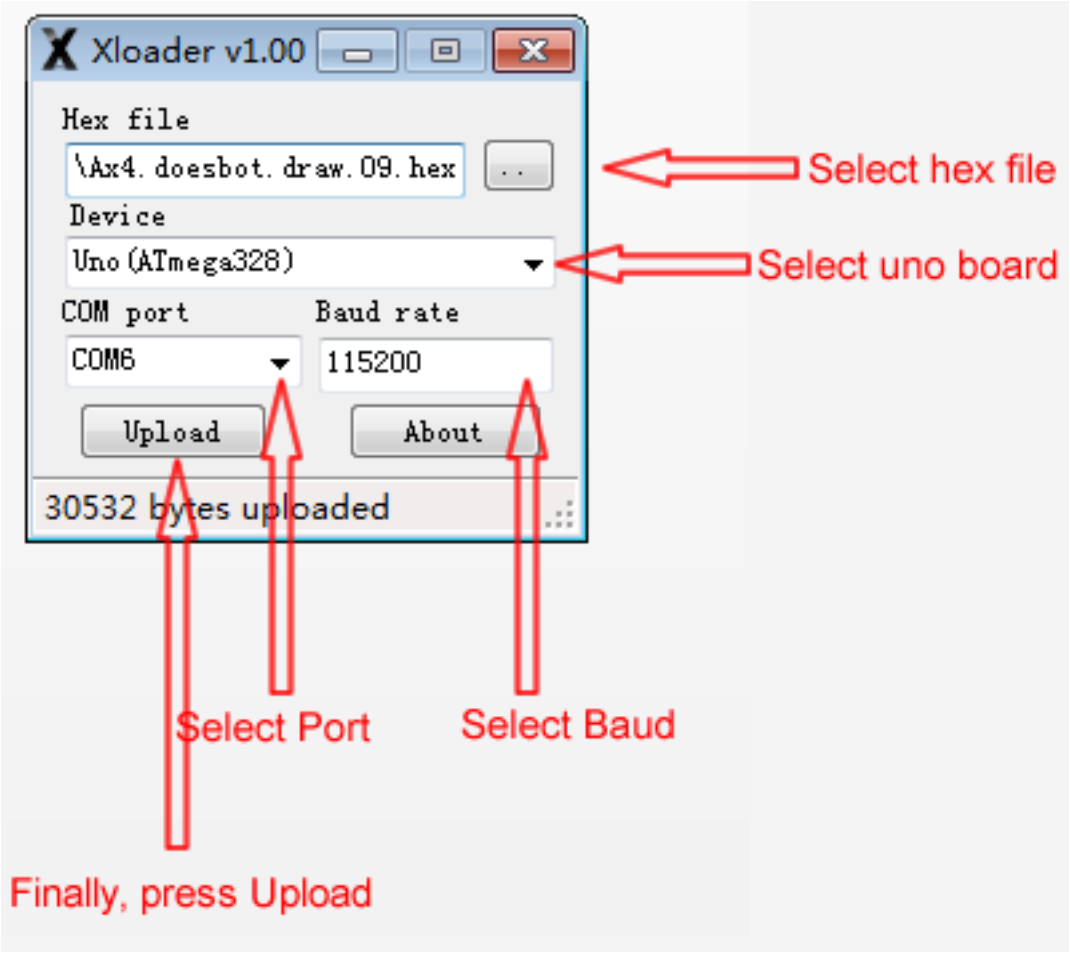


5、Firmware burning

Sometimes we need different versions of firmware, which requires the firmware transmitter Arduino Loader. First, open the directory where the software is located, and then open the software.



Select the corresponding firmware, port, and Arduino type, then press the UPLOAD button at the bottom, wait for a few seconds, and when the word UPLOAD SUCCES appears, it indicates successful upload.



(Please note that the same version of firmware cannot be updated when uploading.

If the current version of the writing machine firmware is 0.9 and you want to burn it, please first upload 1.1.hex in the directory, and then upload the 0.9 version you want to burn.)

In addition to the GRBL0.9 basic software, customers can also use professional micro engraving butlers for laser engraving and Kuixiang engraving for writing. We have a free version, and if you need it, please contact us to request it.

Please indicate the source for reprinting. If there are any errors, please correct them

E-Mail:admin@doesbot.com

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